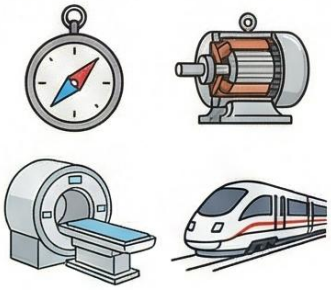
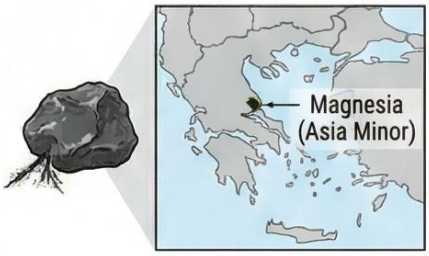
'चुंबकत्व (Magnetism)'

1) Introduction to Magnetism



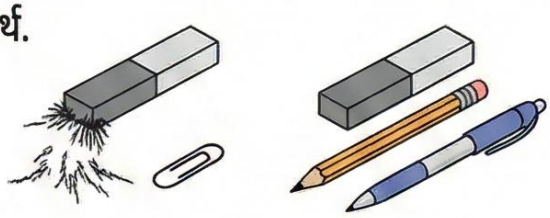
- Magnetism हा आपल्या दैनंदिन जीवनाचा भाग आहे – compass, electric motor, MRI machine आणि Maglev train मध्ये याचा वापर होतो.
- Magnetism ही एक natural physical phenomenon आहे.
- Magnetism आणि electric current यांचा घनिष्ठ संबंध आहे.
- Science आणि modern technology मध्ये Magnetism ला खूप महत्त्व आहे.

2) Discovery of Magnetism



- Natural magnet चा शोध Greece मधील Asia Minor भागातील Magnesia येथे लागला.
- Iron ला आकर्षित करणाऱ्या natural stone ला Magnetite असे म्हणतात.
- Magnetism हा invention नसून discovery आहे.
- 'Magnet' हा शब्द Magnesia या ठिकाणावरून आला आहे.

3) Magnetic Materials

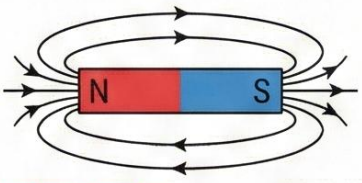


- Magnetic materials म्हणजे magnet कडे आकर्षित होणारे पदार्थ.
- Magnetic materials:
  - Iron (Fe)
  - Nickel (Ni)
  - Cobalt (Co)
- Magnetic materials magnet कडे आकर्षित होतात, तर non-magnetic materials आकर्षित होत नाहीत.

4) Magnet आणि त्याची व्याख्या

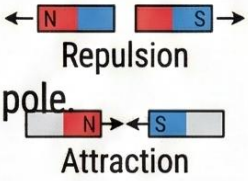
- Magnet म्हणजे असा पदार्थ जो magnetic materials ला आकर्षित करतो आणि magnetic field निर्माण करतो.
- Magnet हा iron ore स्वरूपात सुद्धा आढळतो.
- Iron ores:
  - Magnetite (Fe_3O_4) – only magnetic ore ★
  - Haematite
  - Siderite
  - Limonite

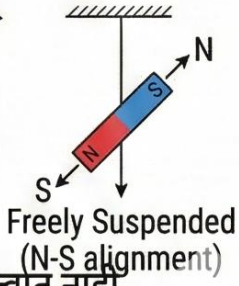
5) Types of Magnets (Exam Focus)



- Natural magnets
- Artificial magnets
- Bar magnet – exam point of view ने सर्वात महत्त्वाचा

6) Properties of Bar Magnet





- प्रत्येक magnet ला दोन poles असतात – North pole आणि South pole.
- Like poles repel each other.
- Unlike poles attract each other.
- Freely suspended magnet नेहमी North–South direction मध्ये स्थिर होतो.
- Magnetic poles नेहमी pairs मध्येच अस्तित्वात असतात; single pole magnet अस्तित्वात नाही.

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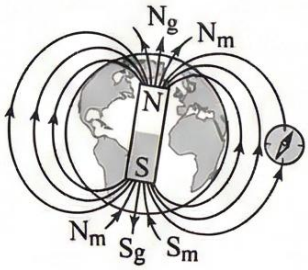
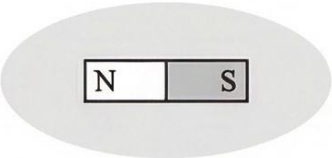
7) Earth's Magnetic Field & Compass



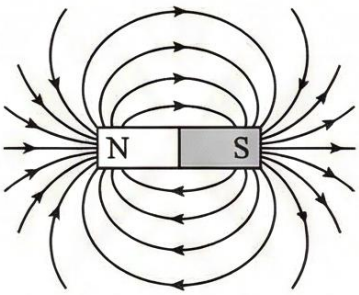
- पृथ्वी ही एक giant magnet प्रमाणे वागते.
- Compass needle नेहमी North–South direction मध्ये स्थिर होते.
- Compass alignment होण्याचं कारण म्हणजे Earth's magnetic field.

8) Magnetic Field



- Magnet च्या आजूबाजूचा ज्या भागात magnetic effect जाणवतो, त्याला magnetic field म्हणतात.
- Magnetic field हा magnetic influence चा region असतो.
- Magnetic field थेट दिसत नाही, पण त्याचा प्रभाव दिसतो.

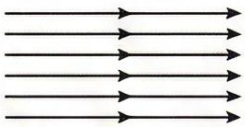
9) Magnetic Lines of Force



- Magnetic lines of force या closed आणि continuous curves असतात.
- Magnet च्या बाहेर या रेषा North pole पासून South pole कडे जातात.
- एखाद्या बिंदूवर काढलेली tangent त्या ठिकाणच्या magnetic field ची direction दाखवते.
- Magnetic field lines कधीच intersect होत नाहीत.
- Magnetic field lines ची density:
  - Crowded lines → Strong magnetic field
  - Sparse lines → Weak magnetic field

10) Types of Magnetic Fields

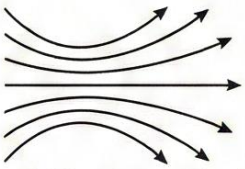
(a) Uniform Magnetic Field



Uniform Magnetic Field

- Magnetic field lines एकमेकांना parallel असतात.
- Magnetic field lines equally spaced असतात.
- Magnetic field ची strength सर्व ठिकाणी same असते.

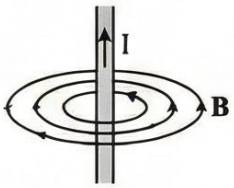
(b) Non-Uniform Magnetic Field



Non-Uniform Magnetic Field

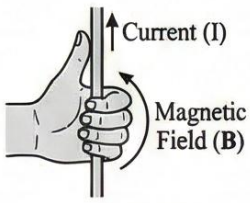
- Magnetic field lines parallel नसतात.
- Magnetic field lines unevenly spaced असतात.
- Magnetic field ची strength ठिकाणानुसार बदलते.

11) Relationship Between Electricity and Magnetism



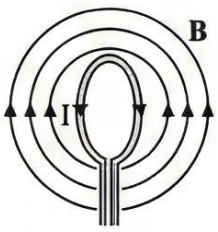
- Hans Christian Oersted यांनी हा संबंध शोधून काढला.
- Current-carrying conductor च्या आजूबाजूला magnetic field तयार होते.
- Electricity आणि Magnetism हे परस्पर संबंधित phenomena आहेत.

12) Right Hand Thumb Rule (Maxwell's Rule)



- Right hand चा thumb → direction of current दाखवतो.
- Curled fingers → direction of magnetic field दाखवतात.
- Straight conductor, circular loop आणि solenoid साठी हा rule वापरला जातो.

13) Magnetic Field Due to Circular Current Loop

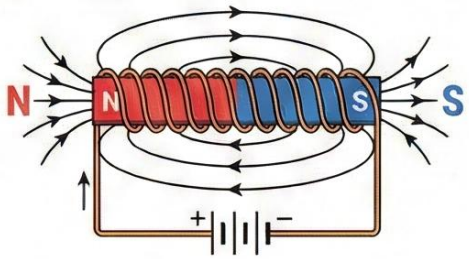


- Circular current loop मधून current प्रवाहित झाला की magnetic field तयार होते.
- Magnetic field concentric circles च्या स्वरूपात दिसते.
- Magnetic field strength अवलंबून असते:
  - Magnitude of current
  - Number of turns

'चुंबकत्व (Magnetism)'

14) Solenoid

- Solenoid म्हणजे insulated copper wire च्या अनेक वळणांनी (turns) बनलेली coil.
- Solenoid चा magnetic field bar magnet सारखा असतो.
- Solenoid हा temporary magnet प्रमाणे वागतो.
- Magnetic field strength खालील घटकांवर अवलंबून असते:
  - Number of turns
  - Current flowing through solenoid
  - Core material / air gap

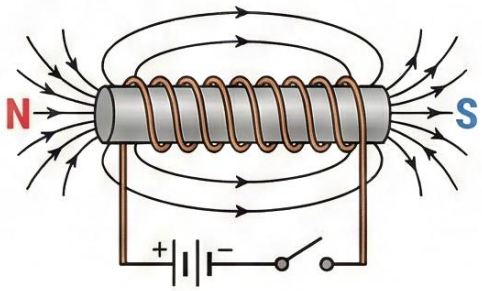


15) Permanent Magnets

- Permanent magnets hard steel वापरून तयार केले जातात.
- वापरले जाणारे materials:
  - Carbon steel
  - Chromium steel
  - Tungsten steel
- Important alloys:
  - Nipermag
  - Alnico
- Applications of permanent magnets:
  - Loudspeakers
  - Microphones
  - Electric meters
  - Speedometers

16) Electromagnet

- Electric current मुळे तयार होणाऱ्या magnet ला electromagnet म्हणतात.
- Construction: solenoid + soft iron core.
- Properties:
  - Temporary magnet
  - Polarity reversible
  - Strength controllable
- Strength खालील घटकांवर अवलंबून असते:
  - Number of turns
  - Current
  - Air gap



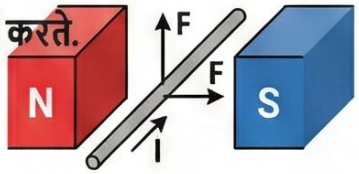
17) Comparison: Bar Magnet vs Solenoid vs Electromagnet

- Exams साठी अतिशय महत्त्वाचा table-based question.

Property	Bar Magnet	Solenoid	Electromagnet
Nature	Permanent	Temporary	Temporary
Source of magnetism	Natural	Electric current	Electric current + soft iron
Polarity	Fixed	Reversible	Reversible
Strength	Constant	Weak	Very strong
Control	Not controllable	Controllable	Highly controllable

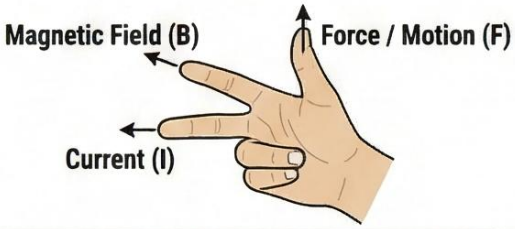
18) Force on a Current-Carrying Conductor

- Magnetic field मध्ये ठेवलेला current-carrying conductor वर force कार्य करते.
- Force अवलंबून असते:
  - Direction of current
  - Direction of magnetic field



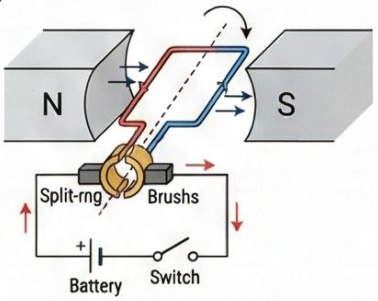
19) Fleming's Left Hand Rule

- Forefinger → Magnetic field ची direction.
- Middle finger → Current ची direction.
- Thumb → Motion किंवा force ची direction.
- Electric motor मध्ये या rule चा वापर होतो.



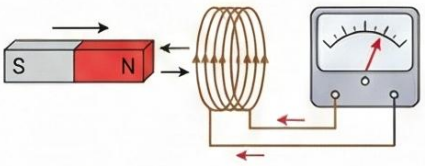
20) Electric Motor

- Electric motor म्हणजे electrical energy ला mechanical energy मध्ये रूपांतर करणारे device.
- Principle: Magnetic field मध्ये current-carrying conductor वर force कार्य करते.
- Construction:
  - Armature coil
  - Strong magnet
  - Split-ring commutator
  - Carbon brushes
  - Battery / DC supply
- Working: Coil मधून current गेल्यावर magnetic force मुळे coil फिरते.
- Applications:
  - Fans, mixers, washing machines
  - EVs, cranes, electric trains



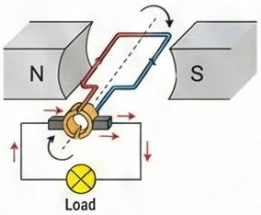
21) Electromagnetic Induction

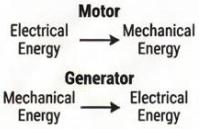
- Michael Faraday यांनी electromagnetic induction चा शोध लावला.
- Changing magnetic field मुळे conductor मध्ये current induce होतो.
- Magnetic field मध्ये conductor च्या motion मुळे current तयार होतो.
- हेच electromagnetic induction चे basic principle आहे.



22) Electric Generator

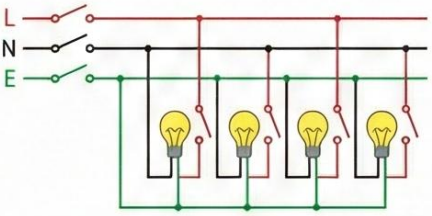
- Electric generator म्हणजे mechanical energy ला electrical energy मध्ये रूपांतर करणारे device.
- Principle: Electromagnetic induction.
- Types of generators:
  - AC Generator
  - DC Generator
- Motor आणि generator या opposite processes आहेत.





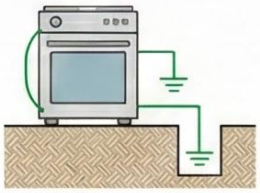
23) Domestic Electric Circuit

- Electricity board कडून घरगुती वीजपुरवठा केला जातो.
- Types of wires:
  - Live wire (Red)
  - Neutral wire (Black)
  - Earth wire (Green)
- House wiring मध्ये appliances parallel connection मध्ये जोडलेले असतात.
- Advantages of parallel circuit:
  - Same voltage for all appliances
  - Independent operation



24) Earthing (Safety)

- Earthing म्हणजे electrical safety साठी earth wire जोडण्याची प्रक्रिया.
- Earth wire leakage current साठी low resistance path देते.
- Electric shock पासून संरक्षण करते.
- "Earthing = Safety First"



25) Overloading

- Overloading म्हणजे circuit मधून excess current जाणे.
- Causes:
  - One socket ला multiple appliances जोडणे
- Effects:
  - Excess current
  - Heating of wires
  - Insulation melting
  - Fire hazards
- Prevention:
  - Proper wire rating वापरणे
  - Excess load टाळणे

26) Applications of Magnetism

- MRI – medical imaging साठी
- Maglev trains – frictionless high-speed transport
- Compass – direction finding
- Galvanometer – current detection

27) Chapter Summary / Revision

- Magnetism आणि electricity यांचा घनिष्ठ संबंध अभ्यासला.
- Solenoid, electromagnet, motor, generator समजले.
- Practical applications आणि safety aspects पाहिले.
- Exam point of view ने महत्त्वाचे concepts revise झाले.